

→ Heuristic Search

8 puzzle → worst case depth = $0 (3^{20})$

↘ 520

✓	×	✓
×	○	×
✓	×	✓

$$\frac{(1 \times 4) + (4 \times 2) + (4 \times 3)}{9} \approx 2.99$$

$$= 3$$

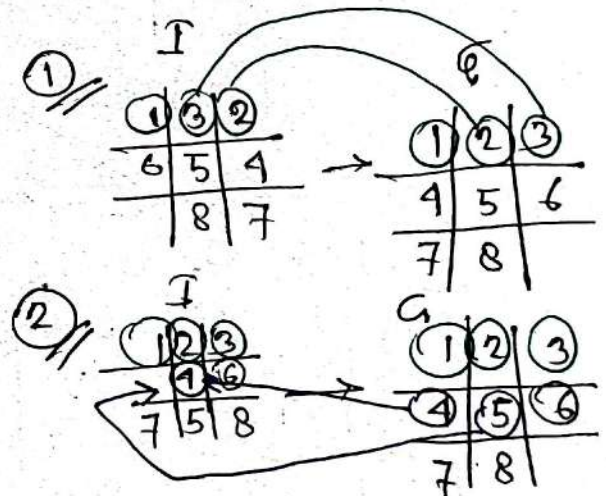
Easy way to solve 8 puzzle problem using heuristic search ↓

I

1	2	3
	4	6
7	5	8

G

1	2	3
4	5	6
7	8	



① Manhattan Distance:

$$0 + 1 + 1 + 2 + 0 + 2 + 0 + 2 = 8$$

② Misplaced Tiles:

$$0 + 0 + 0 + 1 + 1 + 0 + 0 + 1 = 3$$

↪ is misplace? = 1

not misplaced = 0

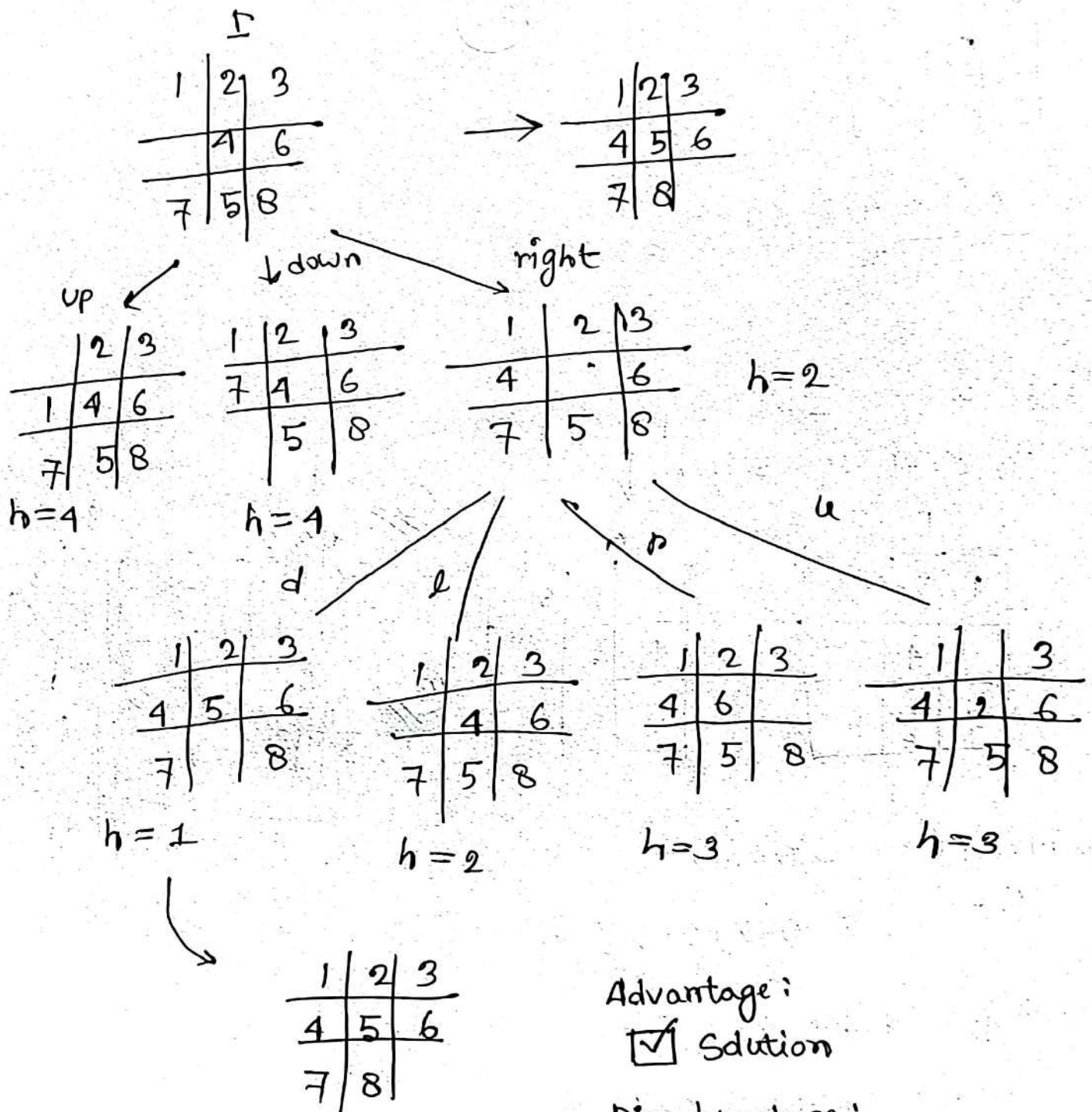
(2)

Q. Solve 8-puzzle using heuristic knowledge

✗ manhattan

✗ misplaced tiles

using misplaced tiles: {Polynomial}



Advantage:

☒ Solution

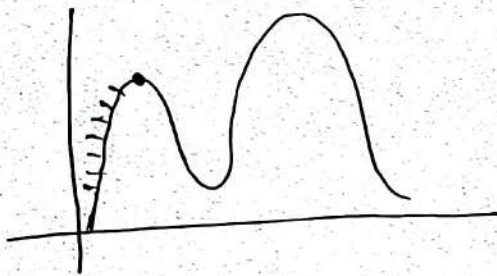
Disadvantage:

☒ Not optimal solution

→ Heuristic Search

① Hill Climbing Search

- ① Simple
- ② Steepest-Ascent
- ③ Simulated Annealing

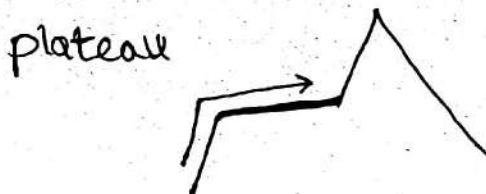
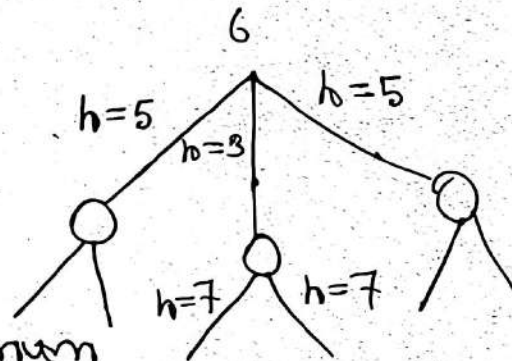


local maximum



ridge

- ☒ Can't change direction
- ☒ Stuck in local maximum
- ☒ Can't reach the goal.



plateau